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Digitální zpracování obrazu

Lecture 4

Potlačení šumu v obraze

- **Modely šumu**
- **Lineární filtrace**
- **Nelineární metody**



Image noise

Apparent in low-light scenes taken at high ISO and/or long exposure



Noise sources

- Background electronic and thermal noise (additive Gaussian)
- Shot noise (photon noise) – Poisson, approx. multiplicative Gaussian
- Random errors of A/D converter, transmission errors (impulse noise)
- Wavefront interference noise (speckle noise)



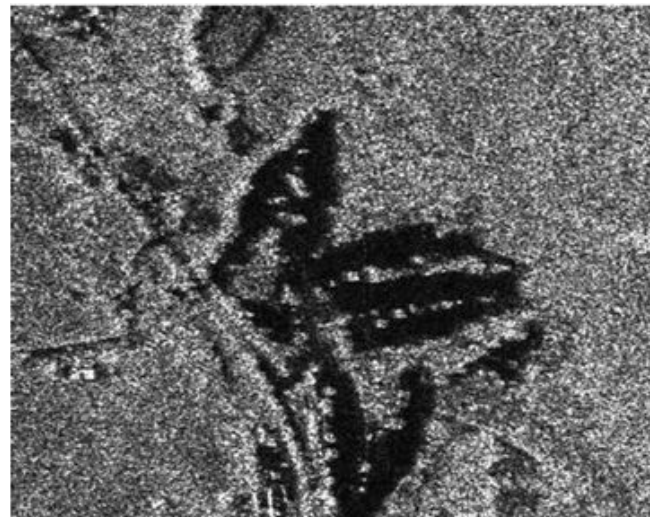
Gaussian



Impulse

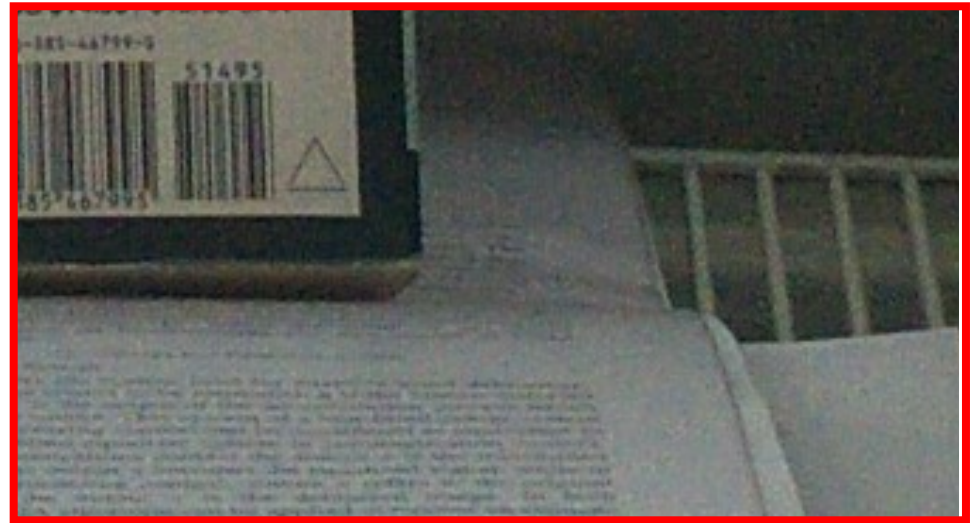


Poisson

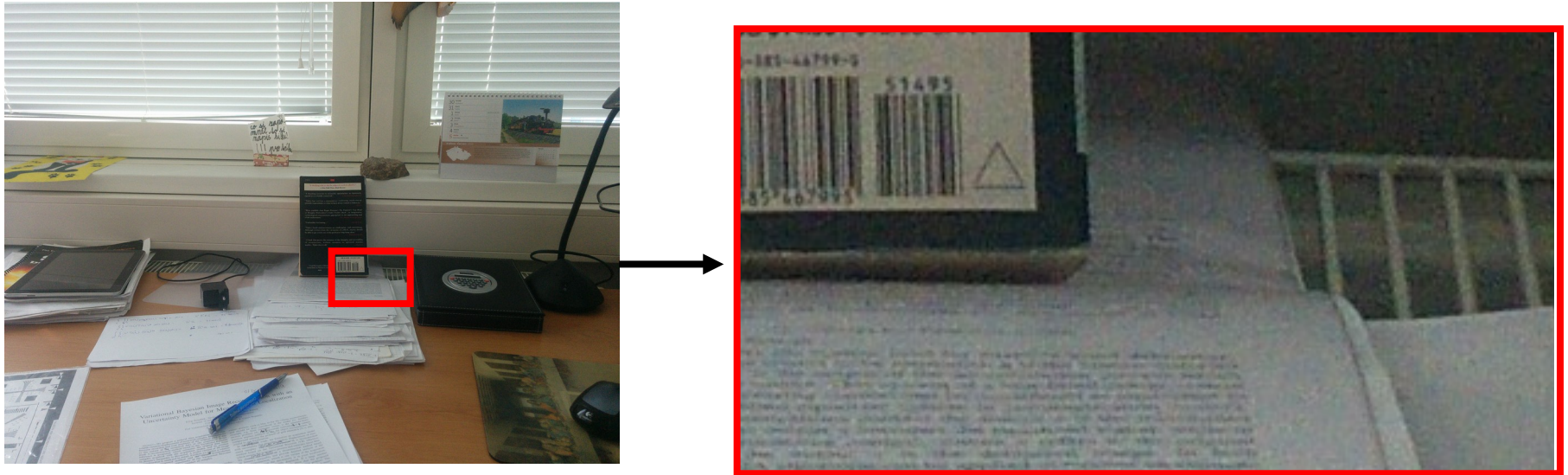


Speckle noise in satellite radar image

Real camera compound noise



Real camera compound noise



Noise models

Non-specific or compound noise is mostly modelled as an additive, signal-independent white Gaussian noise (AWGN)

$$g = \mathcal{D}(f) + n$$

Modely šumu

- **Aditivní náhodný šum**

$$g = f + n$$

- **Gaussovský bílý šum (AGWN)**
- **Impulsní šum (sůl a pepř)**

Modely šumu

- **Gaussovský bílý šum (AGWN)**

- frekvence ?

- korelovanost ?



Modely šumu

- Impulsní šum (sůl a pepř)
- $P(f) = 1 - K$
- $P(1) = K/2$
- $P(0) = K/2$



Míra šumu v obraze

- Rozptyl
- Signal-to-noise ratio (SNR)

$$\text{SNR} = 10 \log (D(f)/D(n)) \quad [\text{dB}]$$

Gaussovský bílý šum

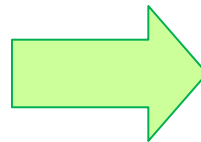


20 dB

10 dB

0 dB

Průměrování v čase



Průměrování v čase

4

8

šum



16

32

64

Průměrování v čase

$g_1, \dots, g_N$ – nezávislá pozorování

$$g_k = f + n_k$$

$$n_k \sim p(0, \sigma^2)$$

$$g = \frac{1}{N} \sum_{k=1}^N g_k = f + n$$

$$n \sim p(0, \sigma^2/N)$$

Průměrování v čase

orig



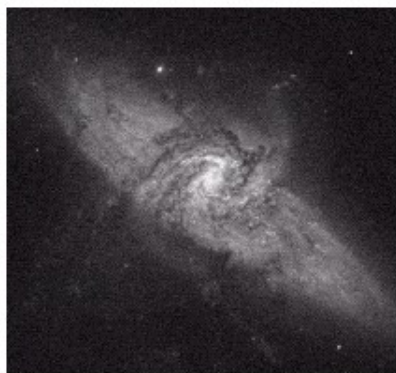
šum



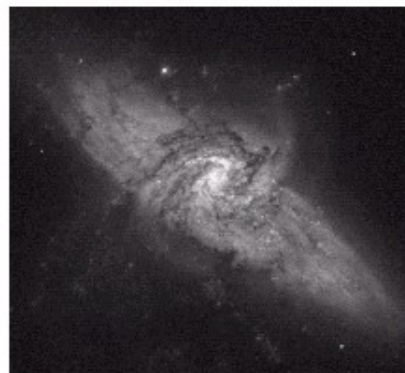
8



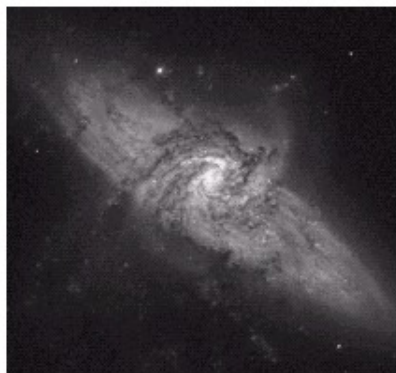
16



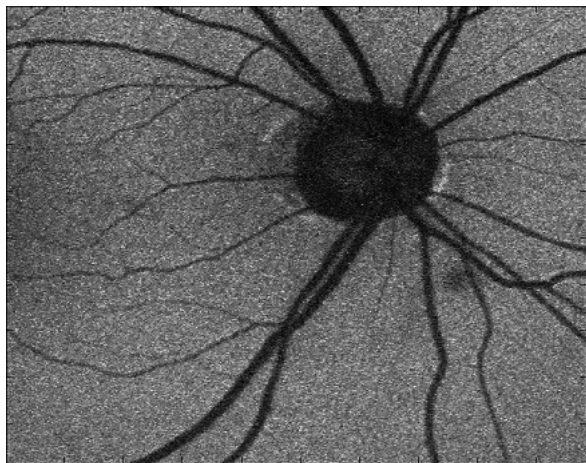
64



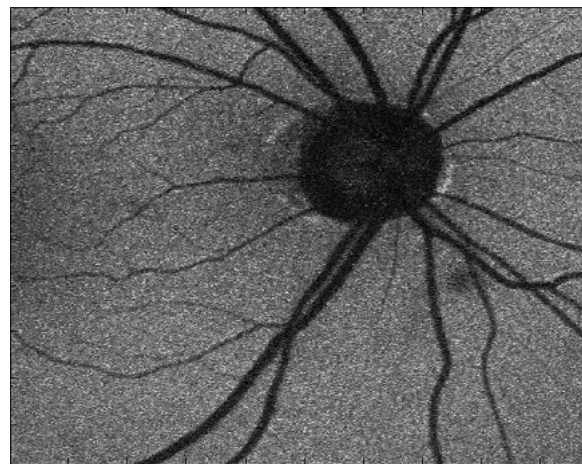
128



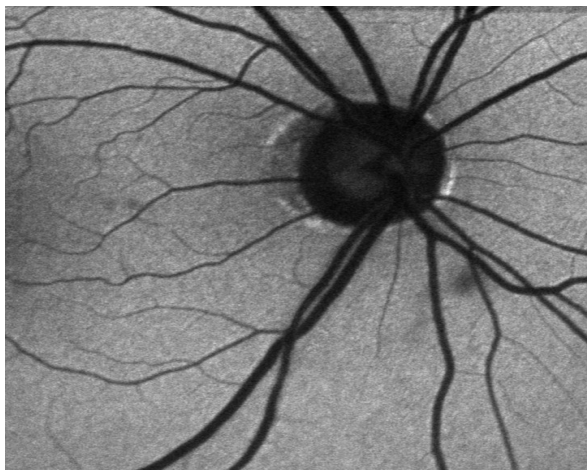
Průměrování v čase



Before registration



After registration



Averaging

Konvoluční filtry

- **Průměrování (prosté a vážené)**
- **Průměrování podél hran**
- **Rotující okno**
- **Filtry ve frekvenční oblasti**

Průměrování v obraze

šum



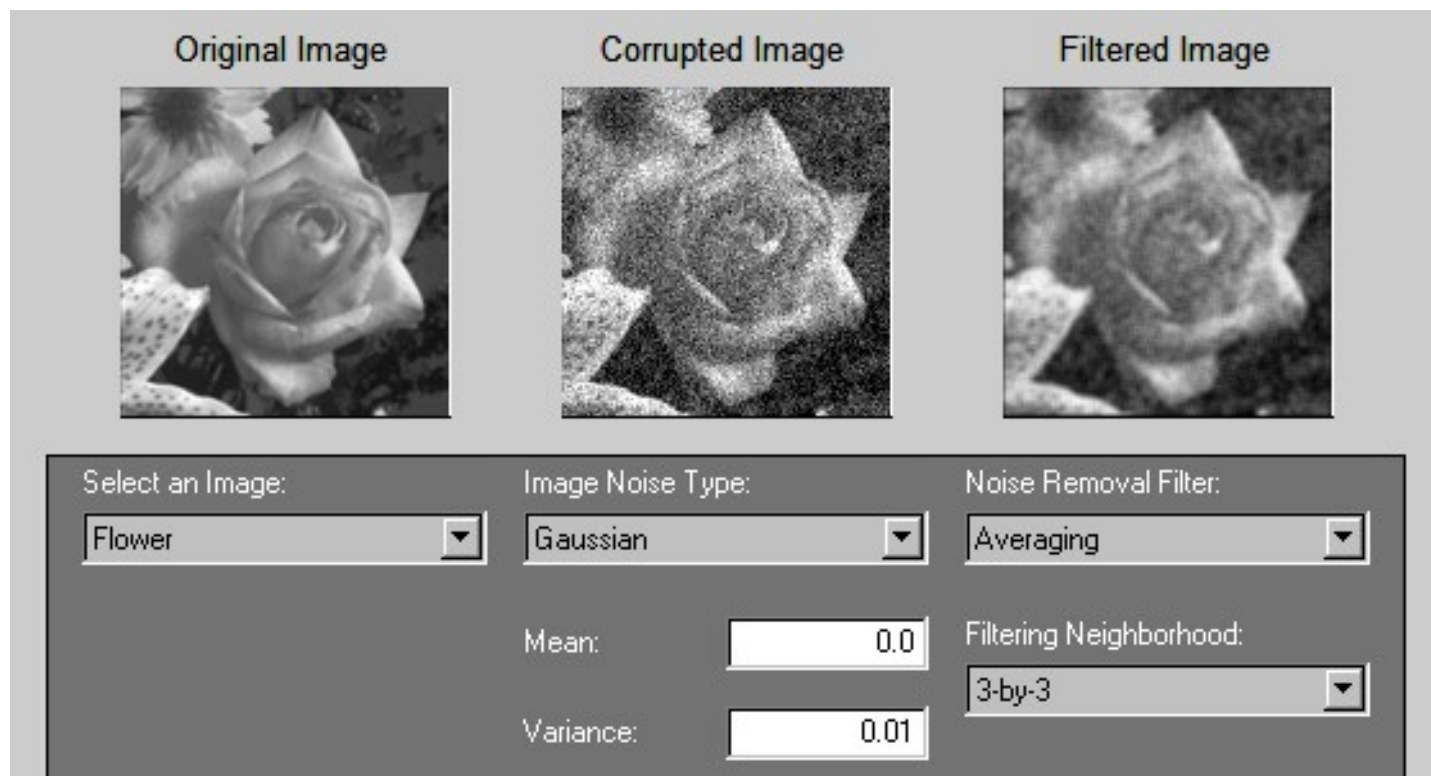
3x3

5x5

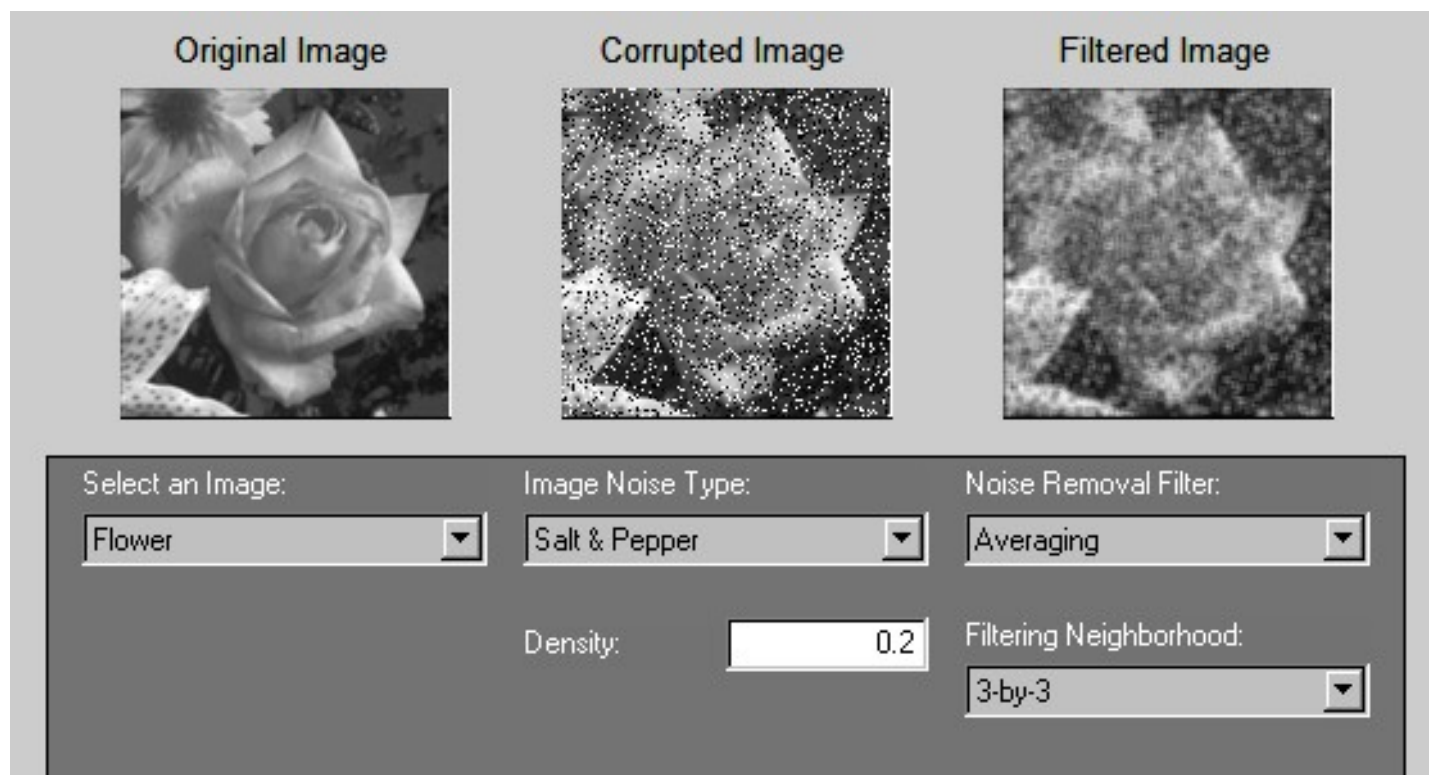


7x7

Průměrování v obraze

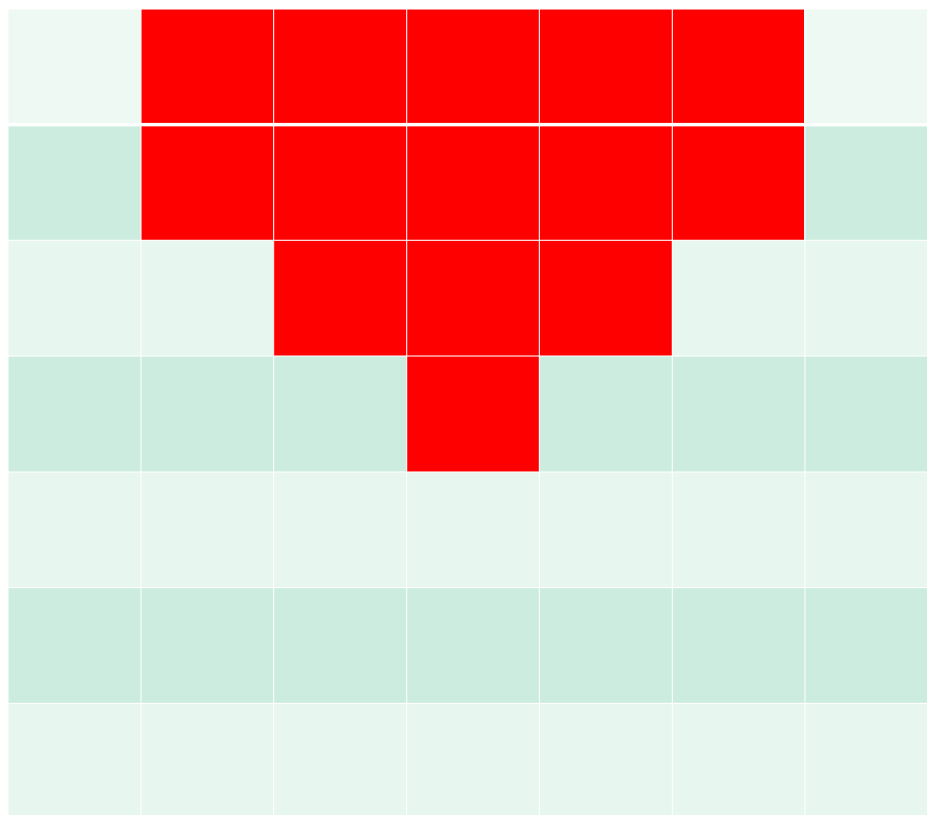


Průměrování v obraze



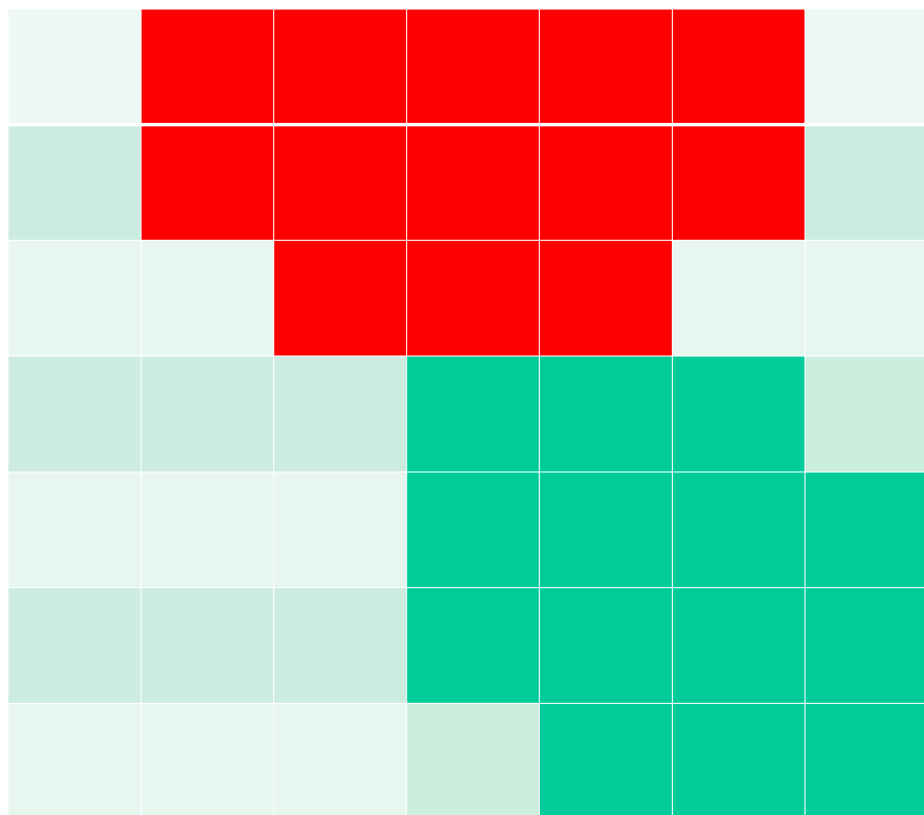
Konvoluční filtry

- Rotující okno

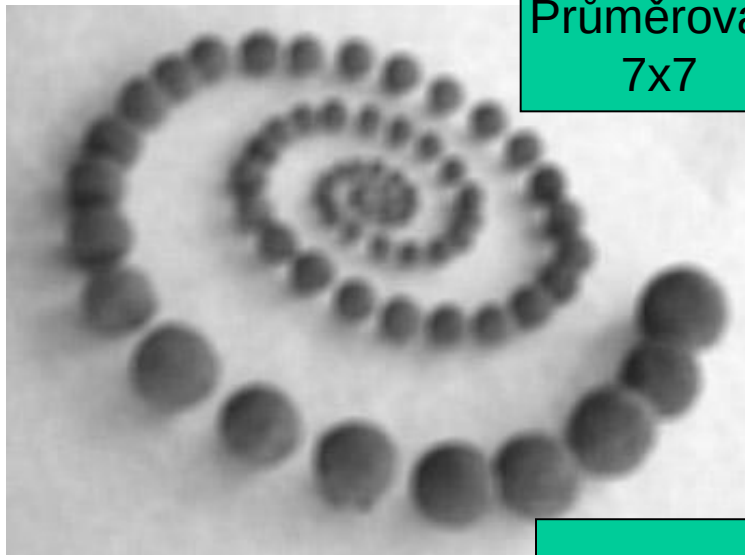


Konvoluční filtry

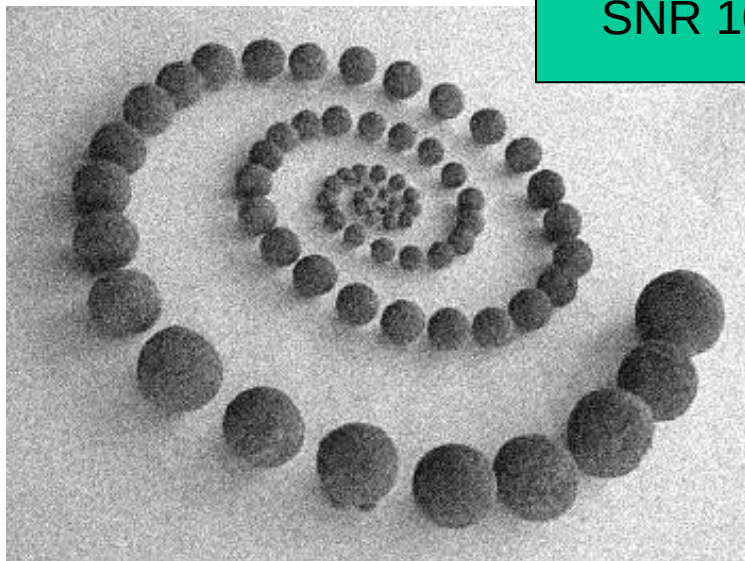
- Rotující okno



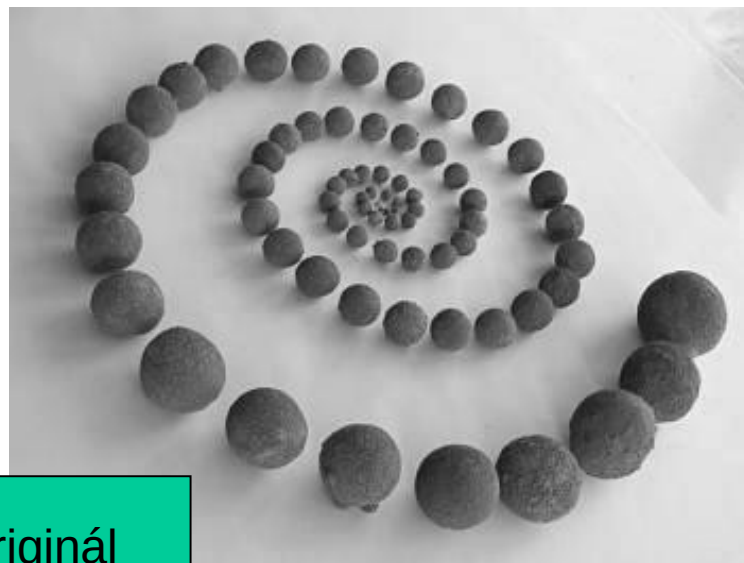
Rotující okno



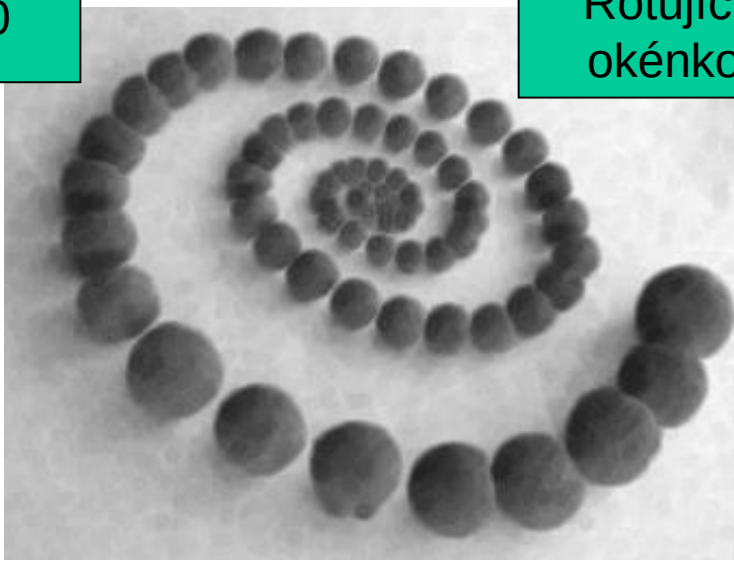
Průměrování
7x7



SNR 10

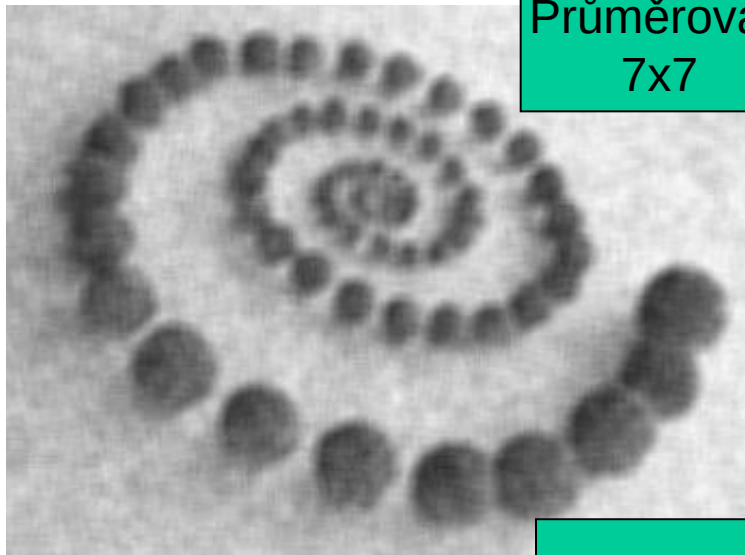


Originál

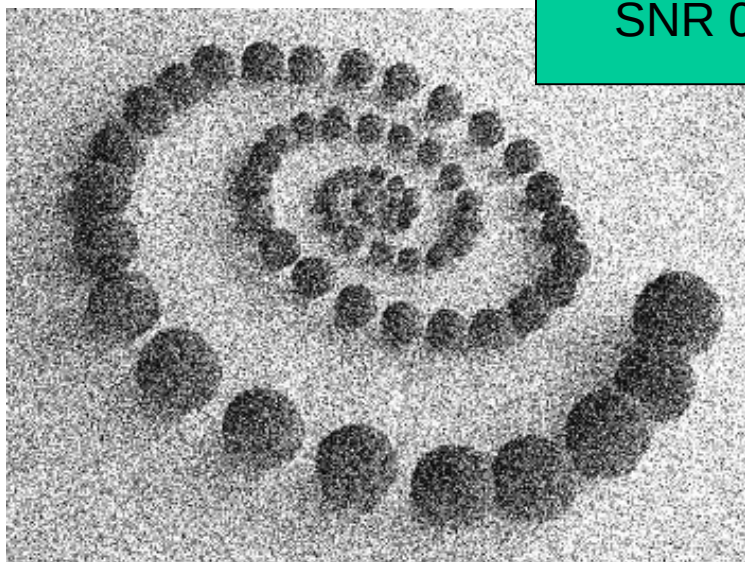


Rotující
okénko

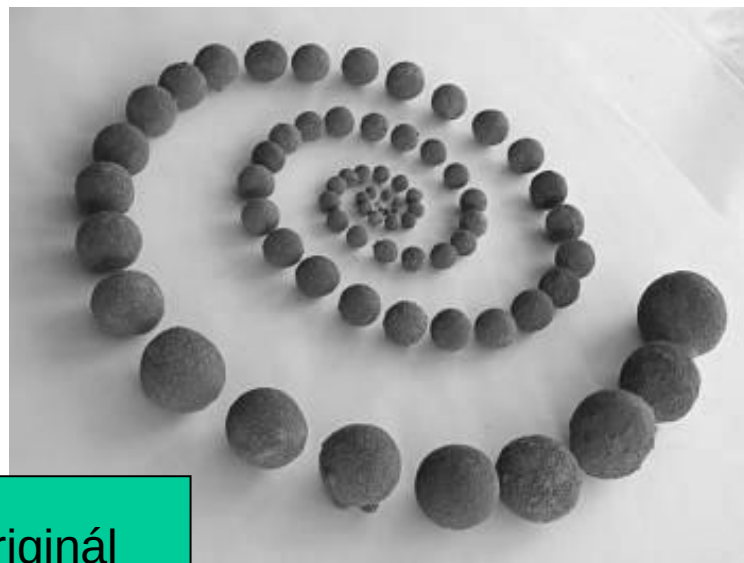
Rotující okno



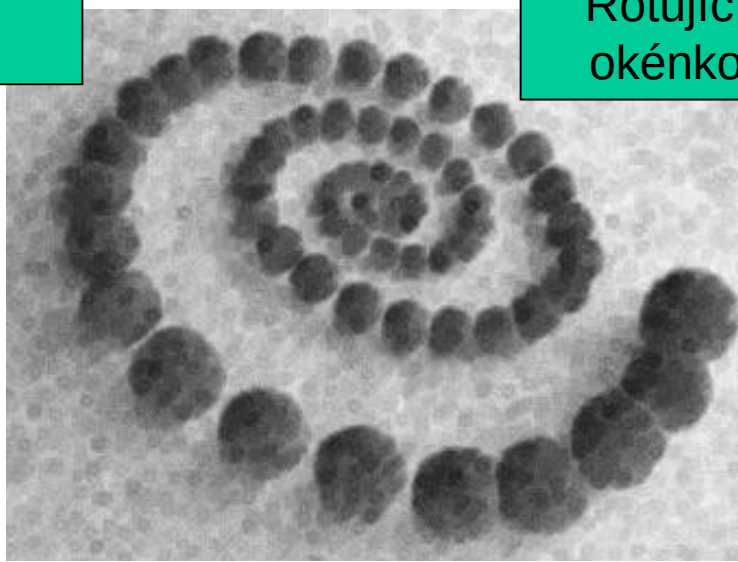
Průměrování
7x7



SNR 0



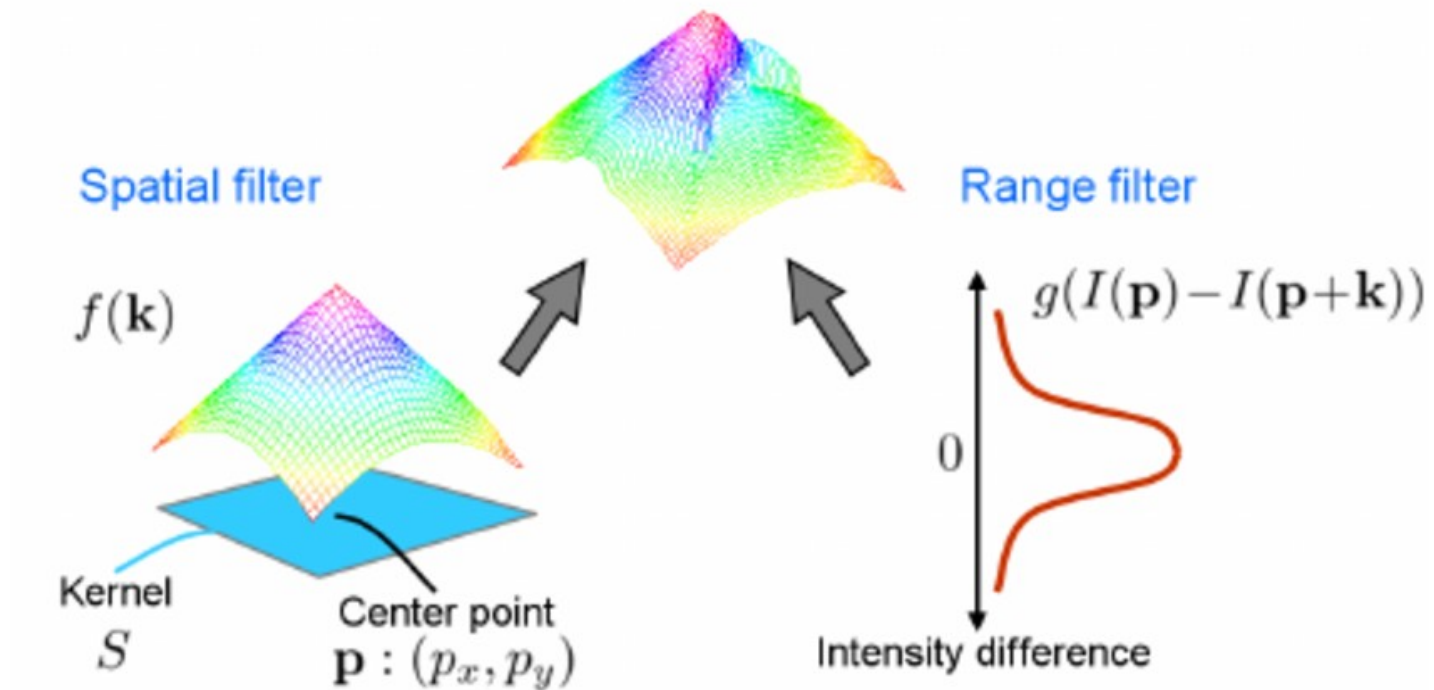
Originál



Rotující
okénko

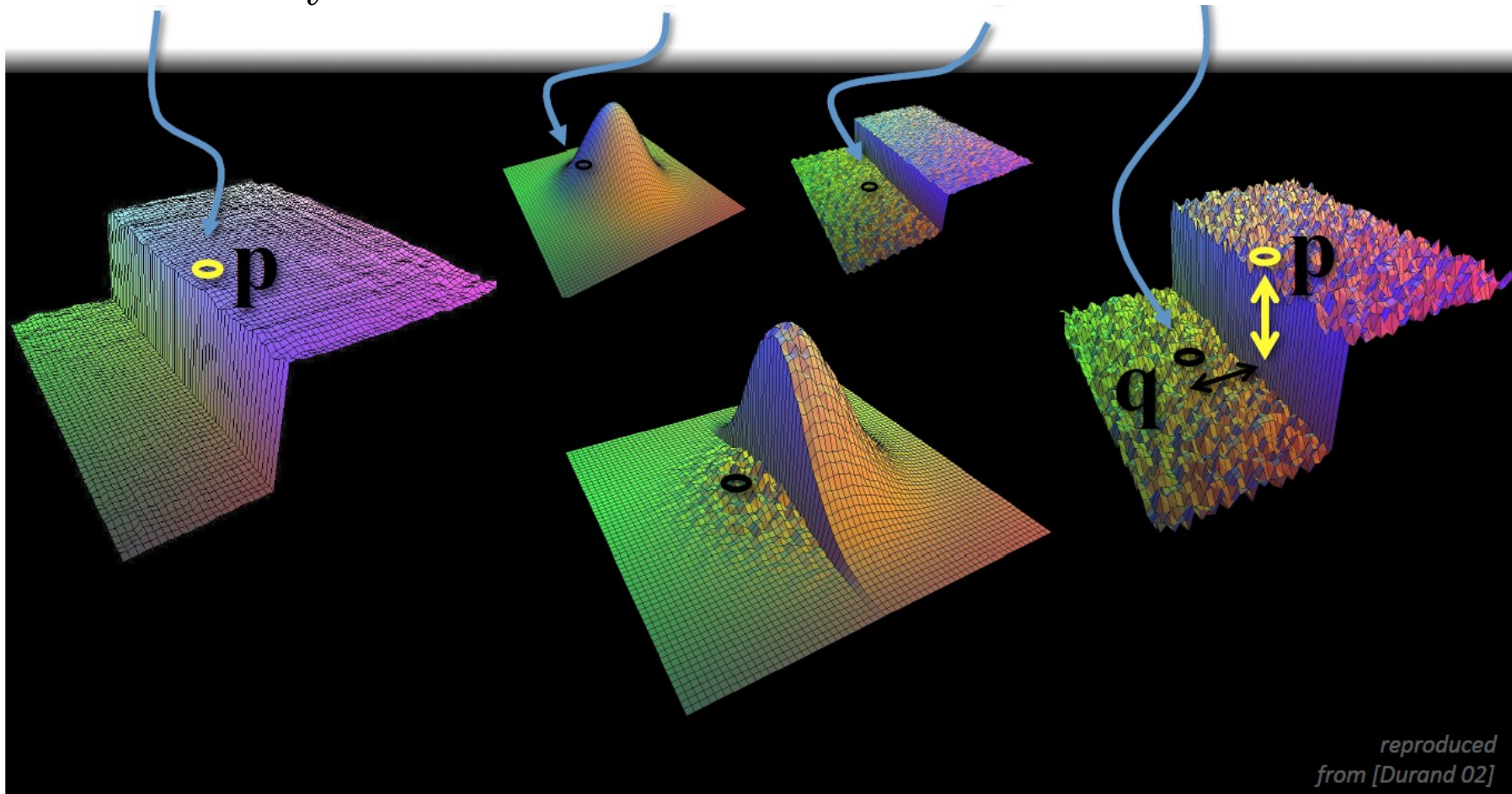
Bilaterální filtr

- Průměrování v okolí přes podobné intenzity (bilaterální filtr)



$$\hat{g}(x) = \sum_t G_{\sigma_s}(\|x - t\|) G_{\sigma_r}(\|g(x) - g(t)\|) g(t)$$

$$\hat{g}(x) = \sum_t \underbrace{G_{\sigma_s}(\|x - t\|)}_{\text{spatial}} \underbrace{G_{\sigma_r}(\|g(x) - g(t)\|)}_{\text{range}} g(t)$$



reproduced
from [Durand 02]



2

0.1



σ_r

0.25



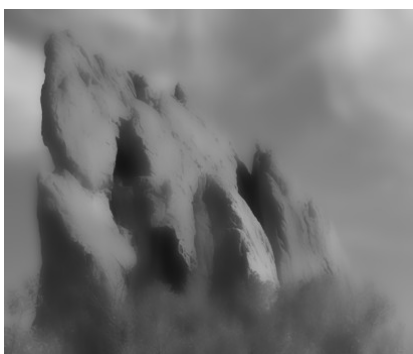
Gaussian blur

∞

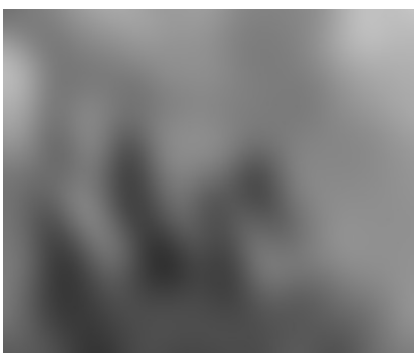
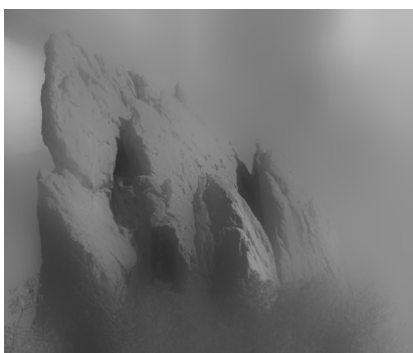


σ_s

6



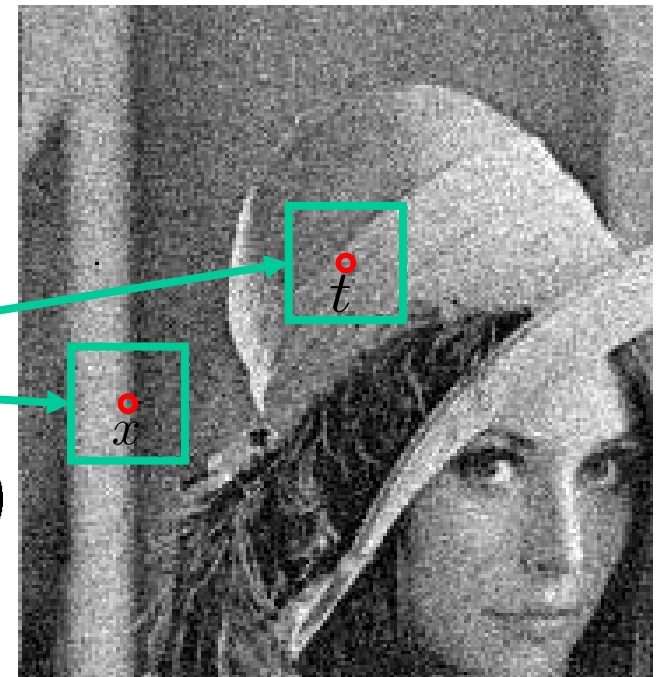
18



Non-local Means filtr

- Průměrování přes všechny pixely s váhou, která je dána podobností okolí

$$\hat{g}(x) = \sum_t \cancel{G_{\sigma_s}(\|x - t\|)} G_{\sigma_r}(\|\mathcal{N}(x) - \mathcal{N}(t)\|^2) g(t)$$



- BM3D (2007)



vstup



**jednoduché
průměrování**



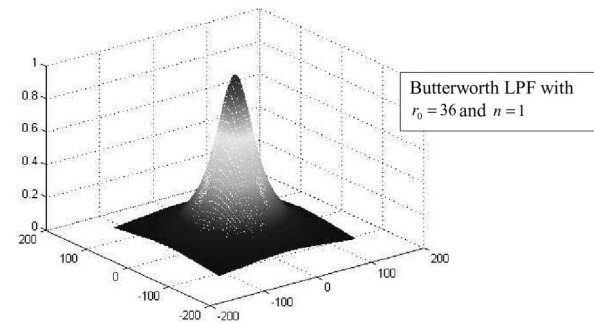
schody

**bilaterální
filtr**



**NL-means
filtr**

Hladký low-pass filtr ve frekvenční oblasti



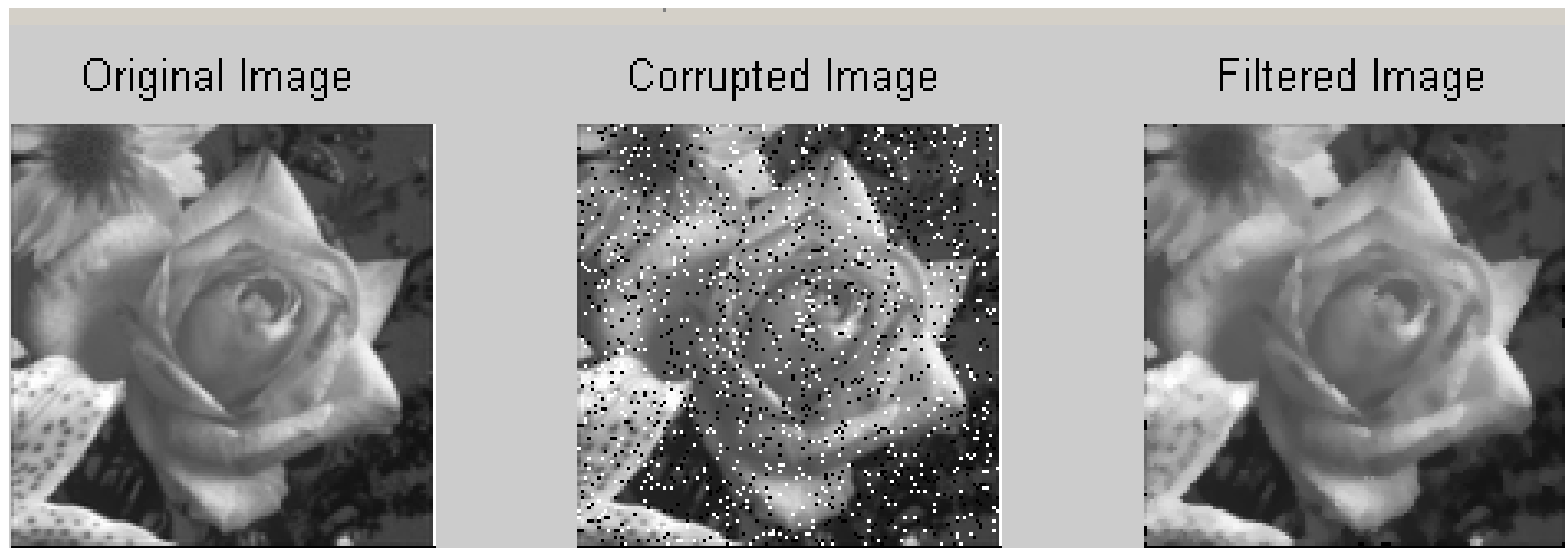
LPF image, $r_0 = 13$



LPF image, $r_0 = 10$

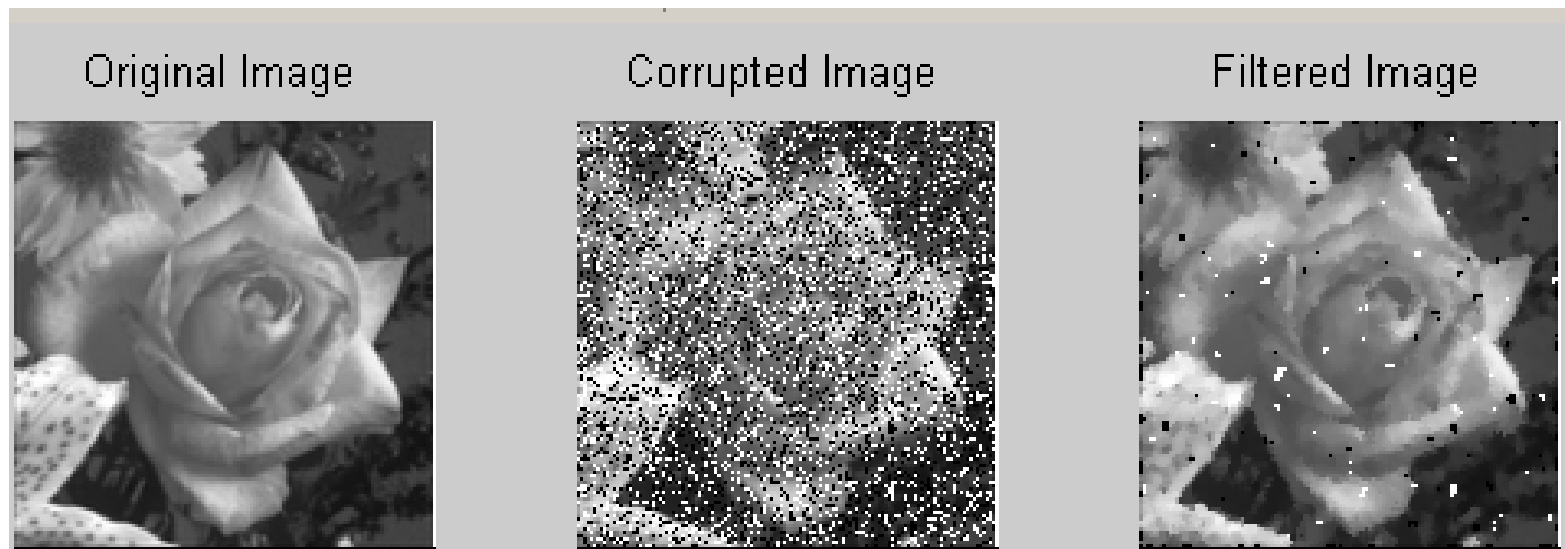
Nelineární filtry

- **Medián**

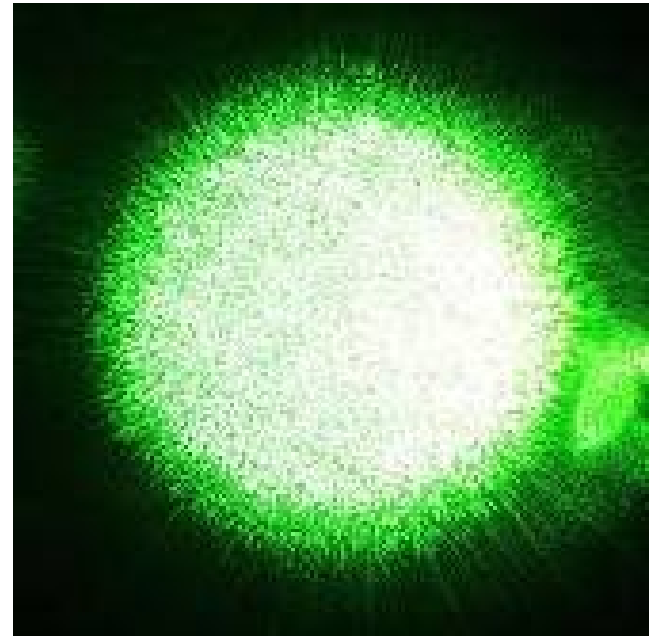
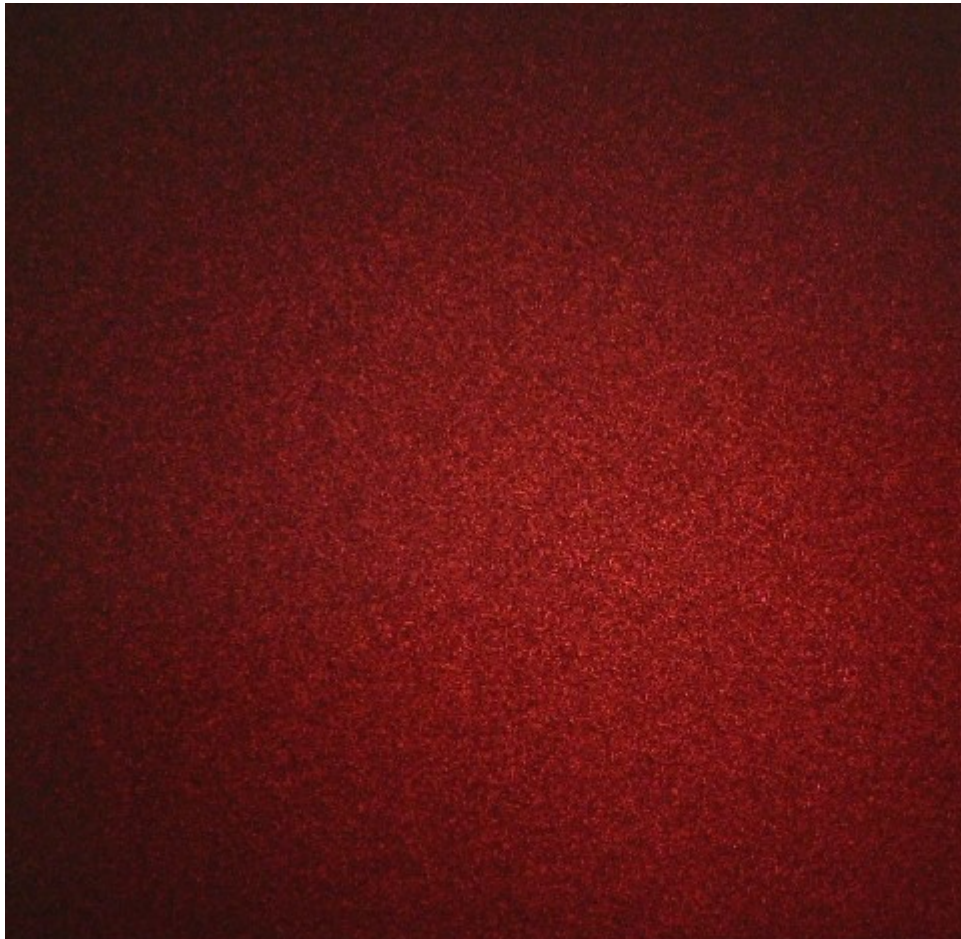


Nelineární filtry

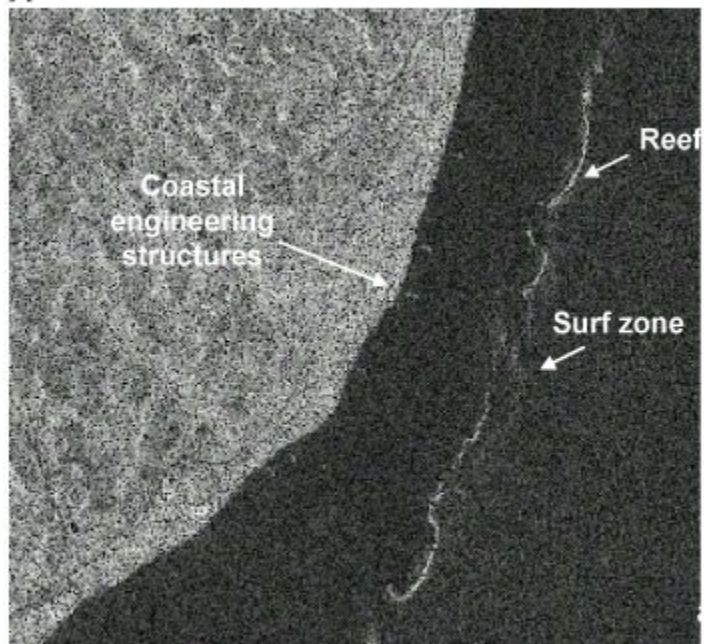
- **Medián**



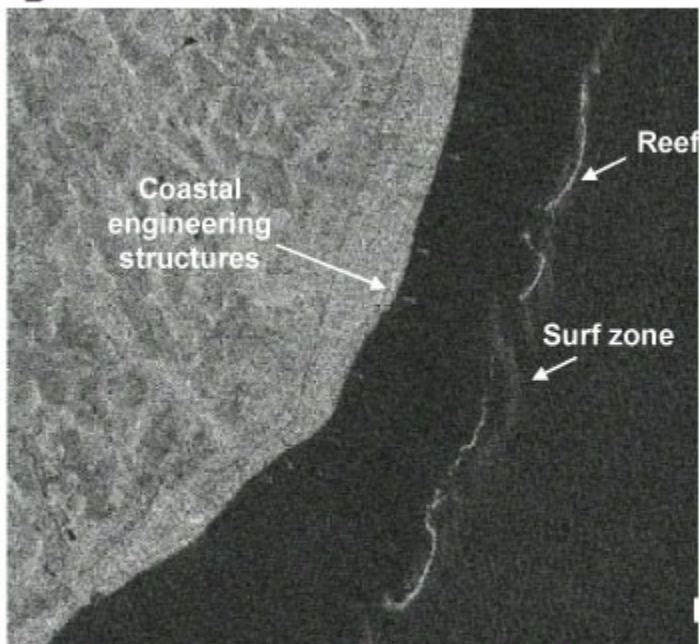
Speckle noise



A



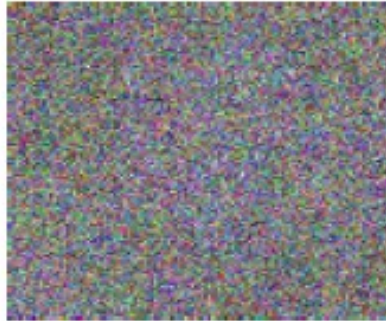
B



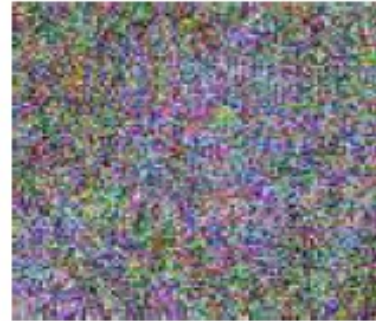
Color median filter



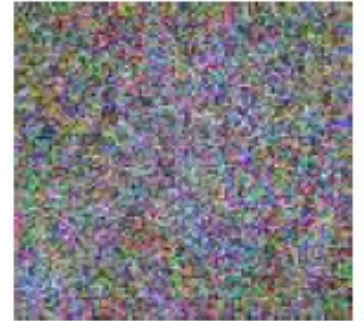
(a)



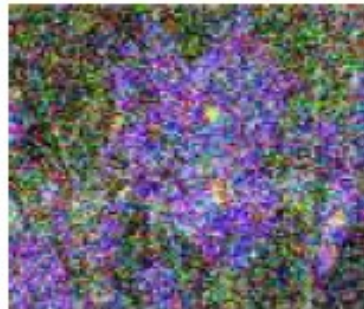
(b)



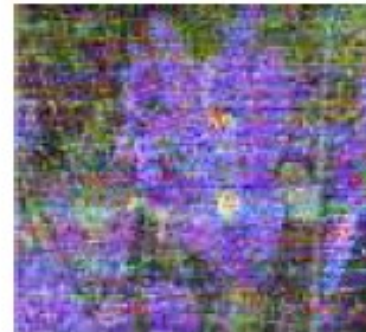
(c)



(d)



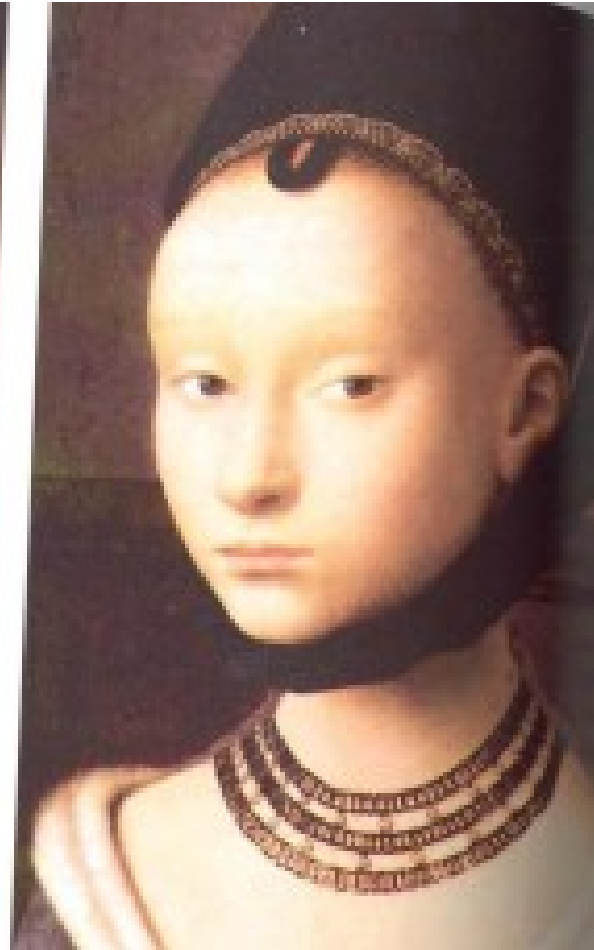
(e)



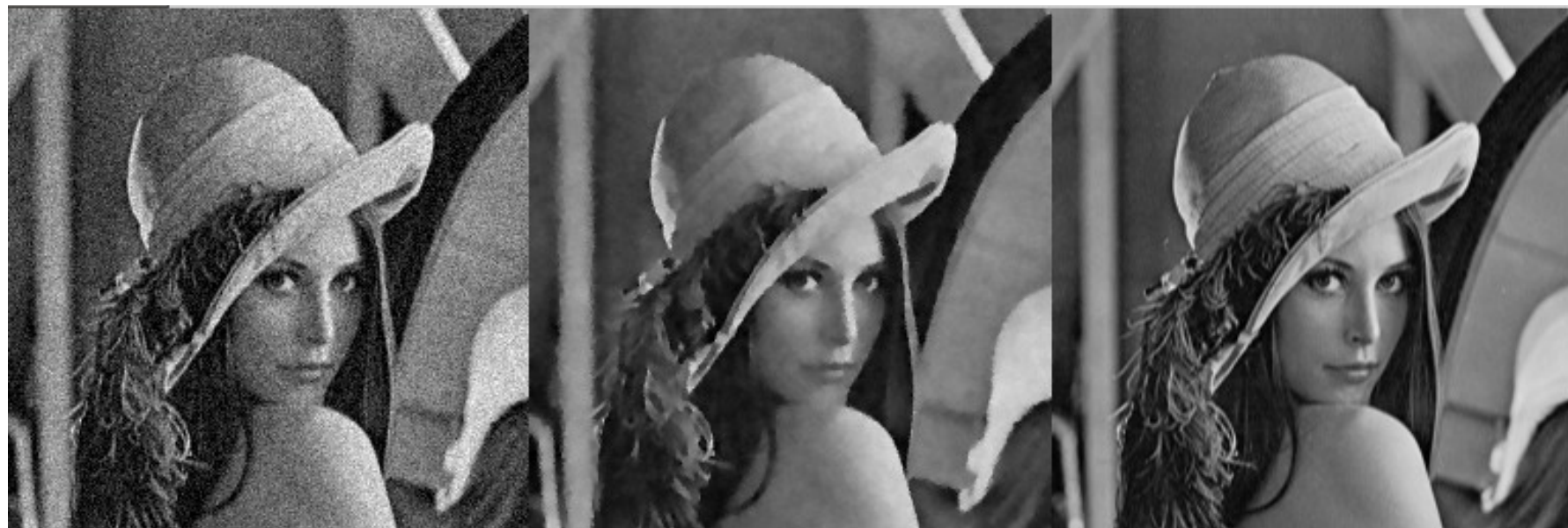
(f)



(g)



Minimalizace funkcionálu



Průměrování v obraze

šum



3x3

5x5

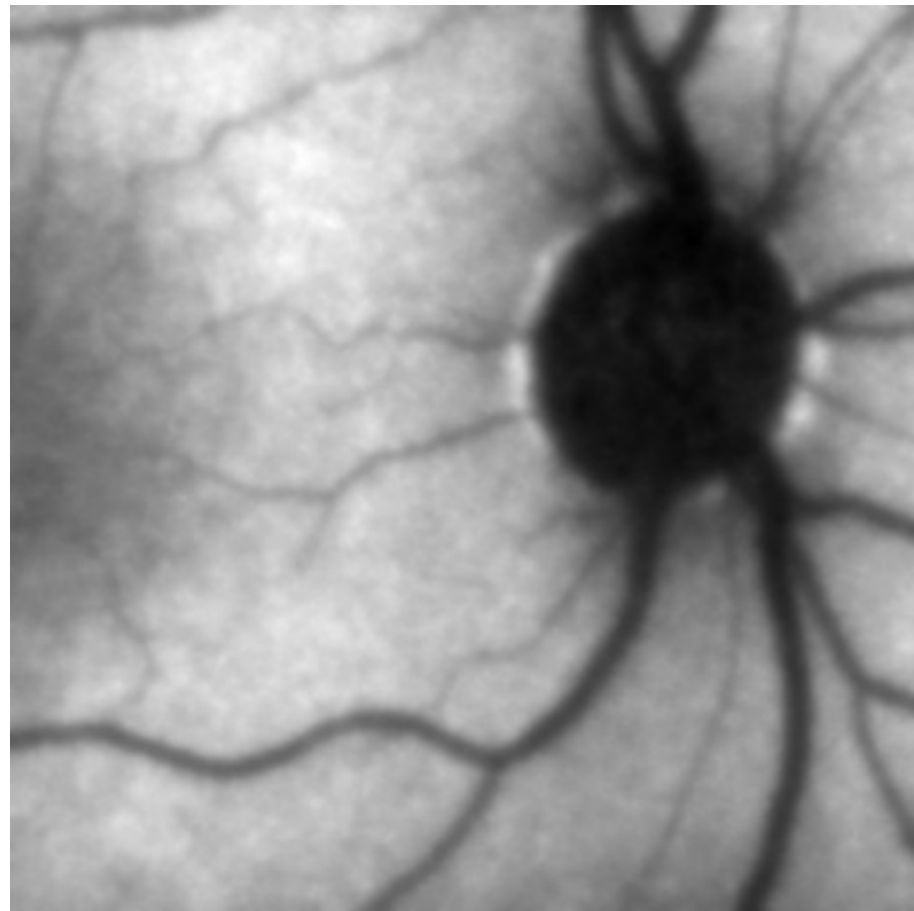
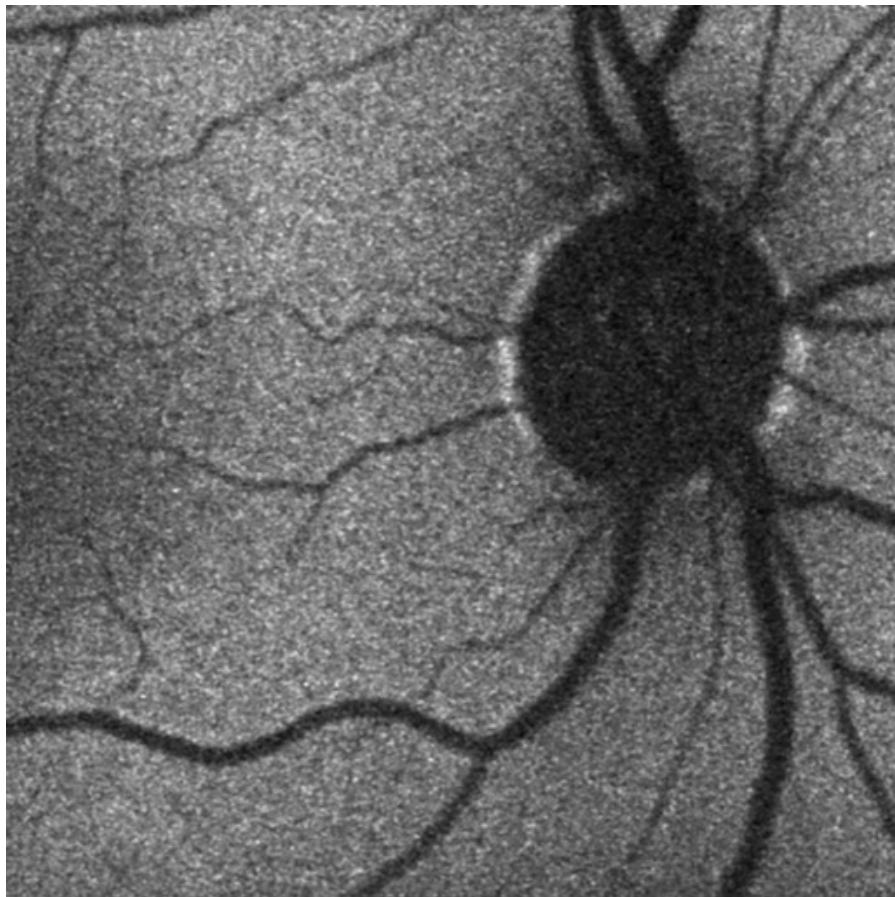


7x7

Infračervený obrázek - splajny



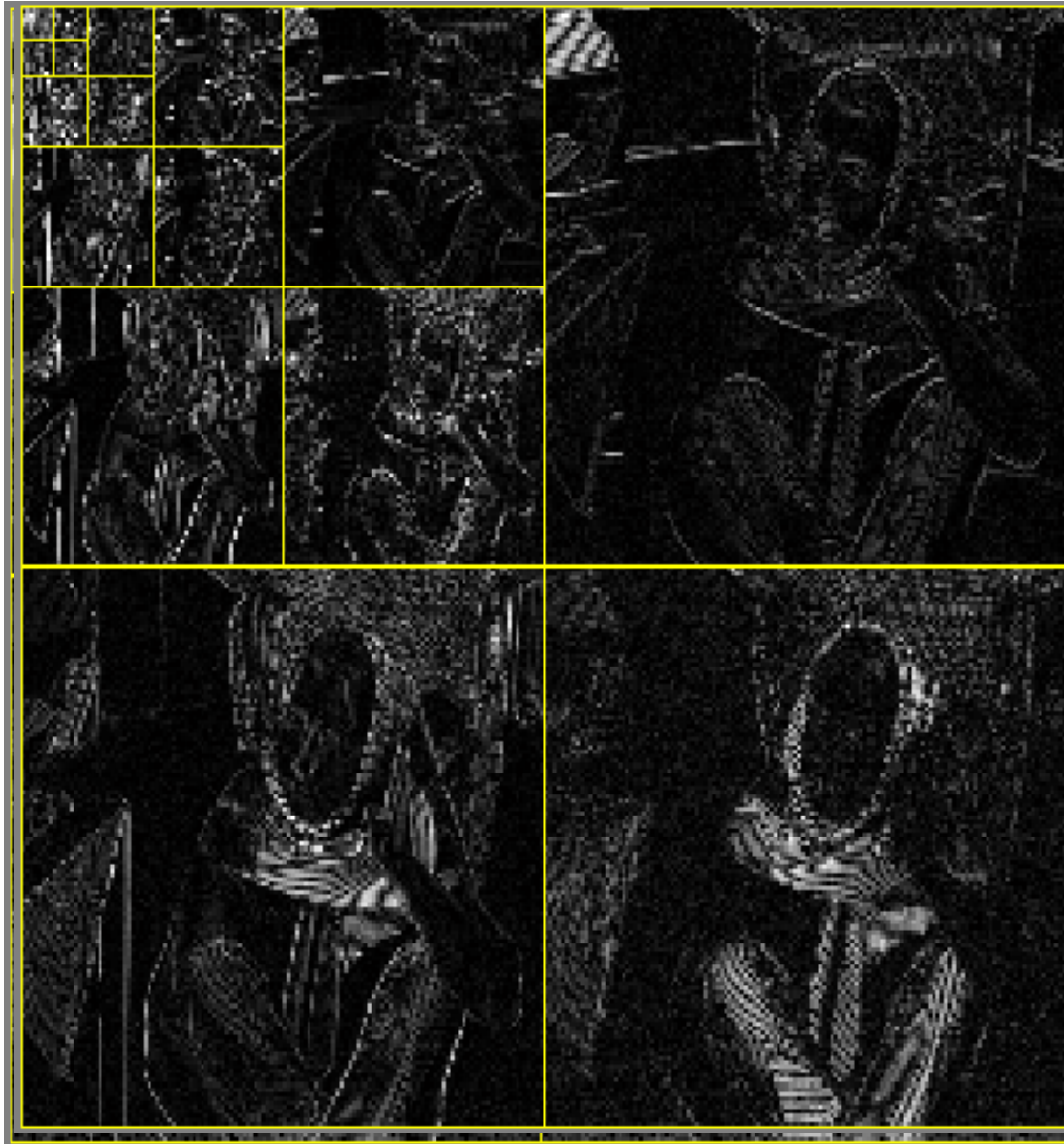
Autofluorescenční snímky oka



Anizotropní difuze

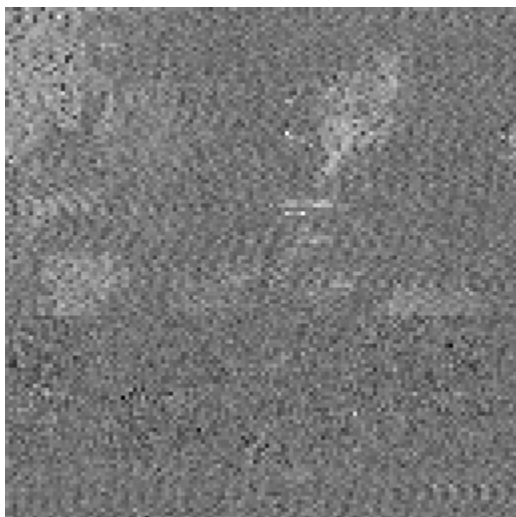


Wavelety

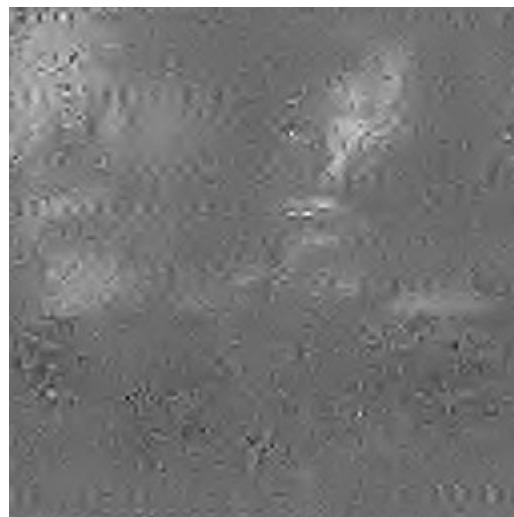


Wavelety

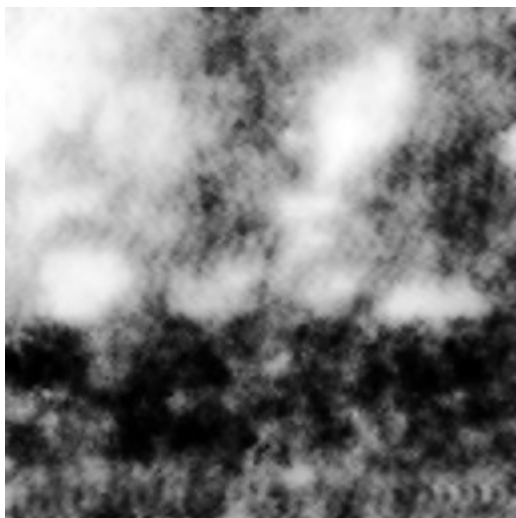




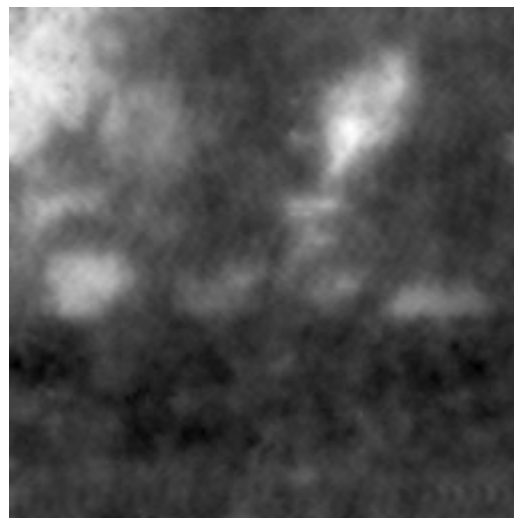
noisy BEEM image



wavelet-based denoising



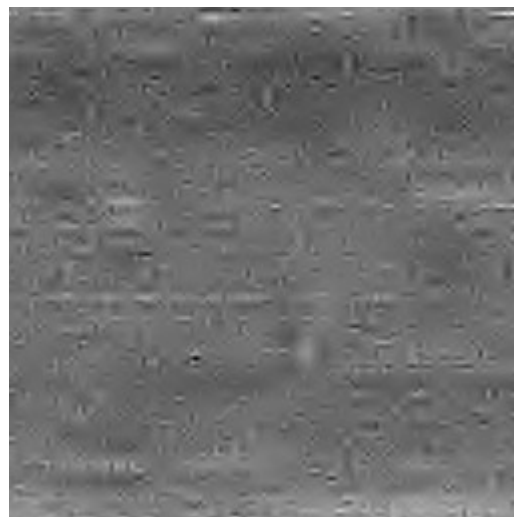
TV + histogram equalization



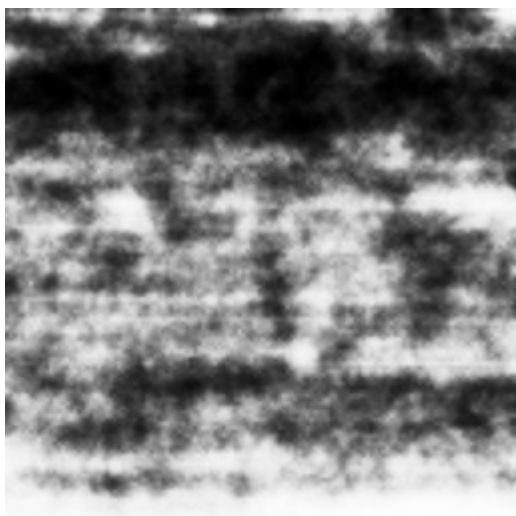
TV-based denoising



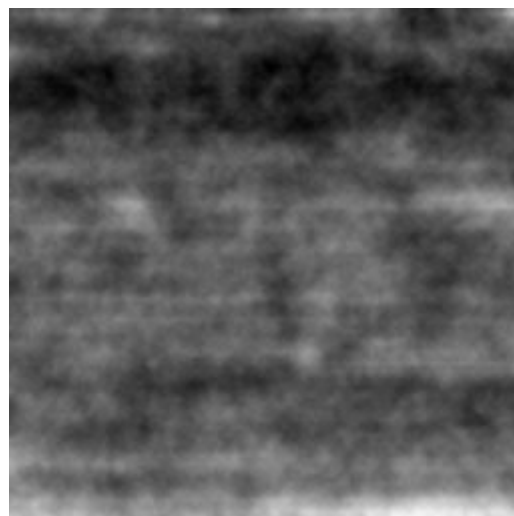
noisy BEEM image



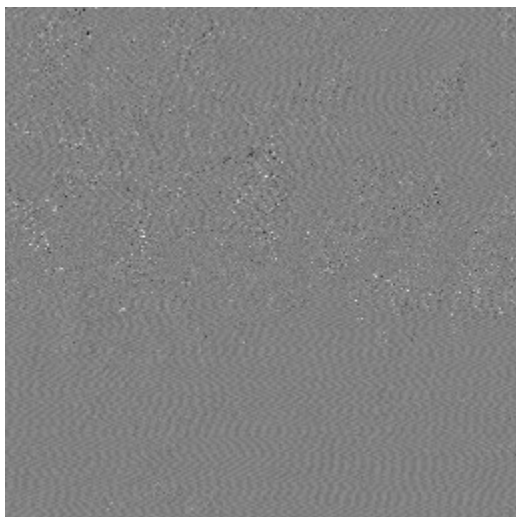
wavelet-based denoising



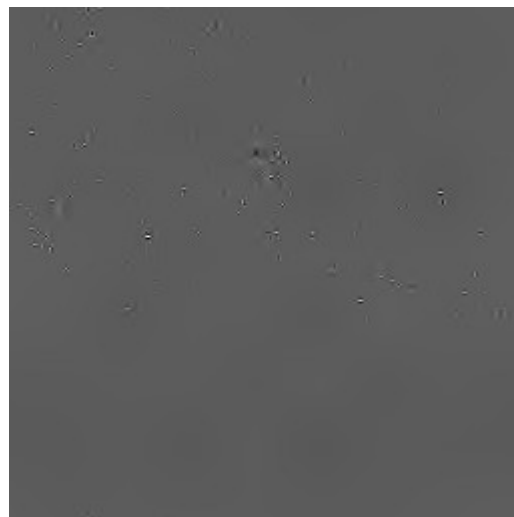
TV + histogram equalization



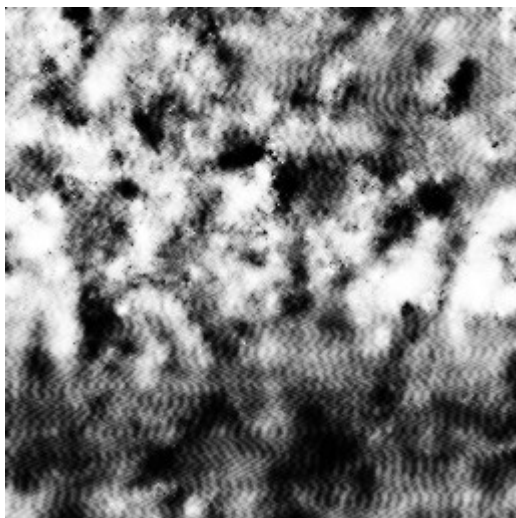
TV-based denoising



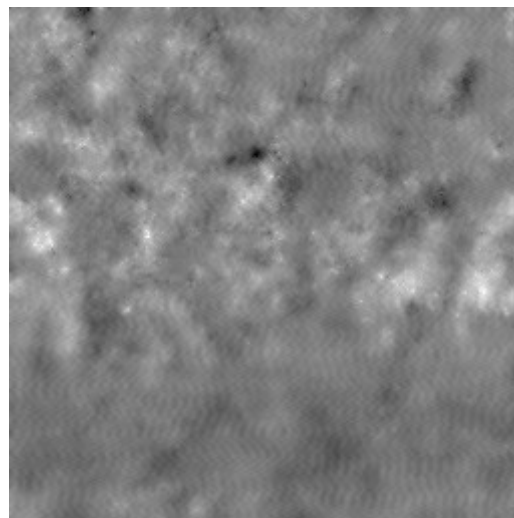
noisy BEEM image



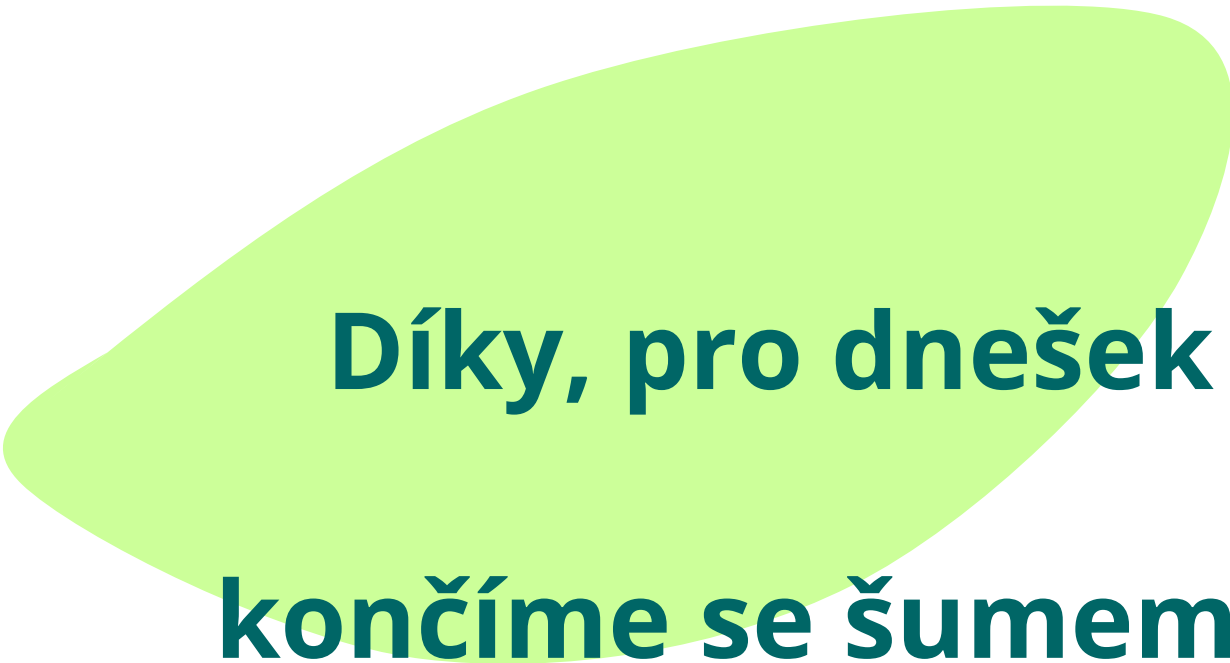
wavelet-based denoising



TV + histogram equalization



TV-based denoising



**Díky, pro dnešek
končíme se šumem!**

Nějaké otázky ?